

JIMIN LEE

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Research interests

- Video Understanding
- Video Representation Learning
- Dense Video Captioning

Education

Sejong University

B.S. in Artificial Intelligence

Total GPA: 3.81/4.5, major GPA: 3.8/4.5

Mar. 2023–Feb. 2027(Expected)

Seoul, South Korea

Experience

Konkuk University, Multimodal AI Lab

Undergraduate researcher

- Advisor: Prof. Jae Won Cho

Feb. 2023–Jul. 2027

Seoul, South Korea

Publications

Stay in your Lane: Role Specific Queries with Overlap Suppression Loss for Dense Video Captioning

Seung Hyup Baek*, **Jimin Lee***, Hyeongkeun Lee, Jae Won Cho (* Equal Contribution)

IEEE International Conference on Computer Vision and Pattern Recognition (CVPR), 2026

- Identified temporal redundancy in localization predictions and repetitive captions as key limitations of query-based dense video captioning models.
- Designed an Overlap Suppression Loss to penalize redundant temporal overlaps while preserving alignment with ground-truth events.
- Developed Cross-Task Contrastive Alignment for decomposed localization and captioning queries and a Concept Guider to enhance the semantic richness of generated captions.

Under review

QR²: Query Relation-Based Reformulation for Dense Video Captioning

Jimin Lee, Seung Hyup Baek, Jae Won Cho

Journal manuscript under review

One Slot, One Moment: Binding Disjoint Scenes and Decoupling Queries in DETR for Video Temporal Grounding

Chanyoung Ko, Seunghyun Mun, **Jimin Lee**, Jae Won Cho
Journal manuscript under review

Projects

Analyzing Digital Transformation Trends Using Multimodal Methods

Korea Information Society Development Institute (KISDI) | May 2025 – Aug. 2025

- Keyword analysis of YouTube videos using an LLM
- Proposed a unified framework that automatically collects videos of various lengths and extracts keywords using an LLM

Personal Projects

Development of a Driver-Aware ADAS Model Using a Vision-Language Model (VLM) |

Mar. 2026–Jun. 2026

Development of a Bedtime Virtual Travel Chatbot Using Prompt Engineering | May 2025

Multimodal Fall Detection and Knowledge Transfer to a Unimodal Model | Sep. 2024–Dec. 2024

Proposal of a Reading Challenge to Promote Reading Participation, Culture, Sports and Tourism Data Utilization Competition | Dec. 2024

Development of a Web Service for Sejong University's Jiphyeon Campus Platform

— Implemented the face-recognition component | Mar. 2024–Jun. 2024

Development of a Message Emotion Classification and Context-Aware Reply Recommendation App |

Sep. 2023–Dec. 2023

The source code for these projects is available on my [GitHub](#).

Skills

Programming Python, C

Deep Learning PyTorch, scikit-learn

Tools Git, Linux, LaTeX

Awards

Excellence Award (KCTI President's Award), Culture, Sports and Tourism Data Utilization Competition

Team: BookTree | Project: A Reading Challenge to Encourage Reading | Dec. 2024